

I: THE ART OF SECRET WRITING

1.1 Introduction to Cryptography Cryptography is the science of using mathematics to encrypt and decrypt data. It is the invisible force that allows us to shop online, send private messages, and use digital signatures without fear of interception. At Duy Tan University, we study the transition from ancient ciphers to the quantum-resistant algorithms of the future.

1.2 The CIA Triad in Cryptography Cryptography is the primary tool used to achieve three core security goals:

- **Confidentiality:** Ensuring only the intended recipient can read the message.
- **Integrity:** Ensuring the message was not altered during transit (using Hashing).
- **Non-repudiation:** Proving that the sender actually sent the message (using Digital Signatures).

1.3 Key Terminology

- **Plaintext:** The original, readable message.
- **Ciphertext:** The scrambled, unreadable version of the message.
- **Cipher:** The mathematical algorithm used for transformation.
- **Key:** A secret piece of information needed to lock or unlock the data.

II: CAREERS IN ENCRYPTION & DATA PRIVACY

2.1 The Architect of Privacy As data breaches become more expensive, companies are hiring specialized experts to ensure their "Crown Jewels" (sensitive data) are encrypted properly.

2.2 Professional Roles

- **Cryptographer:** A mathematical expert who develops new encryption algorithms and protocols.
- **Security Engineer (PKI):** Manages Public Key Infrastructure, including digital certificates and SSL/TLS for websites.
- **Data Protection Officer:** Ensures that encryption policies meet legal requirements like the GDPR or Vietnam's Cybersecurity Law.

2.3 The Future: Quantum Cryptography With the rise of quantum computers, traditional encryption could be broken. Experts who understand **Post-Quantum Cryptography** are currently among the most sought-after professionals in the defense and banking sectors.

III: SYMMETRIC VS. ASYMMETRIC ENCRYPTION

3.1 Symmetric Encryption (One Key) The sender and receiver use the same secret key to encrypt and decrypt. It is very fast and efficient for large amounts of data.

- **Common Algorithm:** AES (Advanced Encryption Standard) – used by governments and the military worldwide.
- **Challenge:** How do you securely share the secret key with the receiver?

3.2 Asymmetric Encryption (Two Keys) Uses a **Public Key** (to encrypt) and a **Private Key** (to decrypt).

- **Common Algorithm:** RSA and ECC (Elliptic Curve Cryptography).
- **Benefit:** You can share your Public Key with the whole world, and only *you* can unlock the messages sent to you.

3.3 Hashing: The Digital Fingerprint Unlike encryption, hashing is one-way. It turns data into a fixed-length string (e.g., **SHA-256**). If even one character of the original file changes, the hash changes completely. This is how we verify that a file has not been infected with a virus.

IV: DEEP DIVE: SSL/TLS & THE WEB OF TRUST

4.1 How HTTPS Works When you see the padlock icon in your browser, you are using **TLS (Transport Layer Security)**. It is a "Hybrid" system:

1. Asymmetric encryption is used to securely exchange a secret key.
2. Symmetric encryption is then used to encrypt the actual data (for speed).

4.2 Digital Certificates A **Certificate Authority (CA)** acts like a digital notary, vouching that a website (like duytan.edu.vn) is who they say they are.

4.3 Practical Lab Exercise

1. **Environment:** Students use **OpenSSL** (a command-line tool).
2. **Task 1:** Create a pair of RSA keys (Public and Private).
3. **Task 2:** Encrypt a text file using a classmate's Public Key.
4. **Task 3:** Generate a Hash (MD5 or SHA-256) of a file and then change one letter to see the "Avalanche Effect."

4.4 Conclusion

Cryptography is the ultimate shield. By mastering these mathematical tools, DTU students learn how to make data useless to hackers, even if they manage to steal it.